ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

# ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

## PRACTICE EXAM 2 · MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central NC(V) Level 2 Electronics and Digital Concepts for Robotics practice exam 2.**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

**How to mark:** each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable sketches.

## SECTION A

## QUESTION 1

**1.1** 1.1.1 D ✓ 1.1.2 B ✓ 1.1.3 C ✓ 1.1.4 A ✓ 1.1.5 B ✓ 1.1.6 D ✓ 1.1.7 A ✓ 1.1.8 C ✓ 1.1.9 C ✓ 1.1.10 A ✓ (10)

**1.2** 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 True ✓ 1.2.4 False ✓ 1.2.5 True ✓ 1.2.6 True ✓ 1.2.7 True ✓ 1.2.8 False ✓ 1.2.9 True ✓ 1.2.10 False ✓ (10)

**1.3** 1.3.1 J ✓ 1.3.2 G ✓ 1.3.3 D ✓ 1.3.4 A ✓ 1.3.5 C ✓ 1.3.6 I ✓ 1.3.7 E ✓ 1.3.8 B ✓ 1.3.9 F ✓ 1.3.10 H ✓ (10)

**TOTAL SECTION A: [30]**

## SECTION B

## QUESTION 2

**2.1** (a) 2,2 kΩ ✓ (b) 45 mA ✓ (c) 3 300 000 Ω ✓ (d) 0,015 F ✓ (4)

**2.2** Nucleus labelled (14 protons and 14 neutrons) ✓; first shell 2 electrons ✓; second shell 8 electrons ✓; outer (valence) shell 4 electrons ✓ (4)

**2.3** Copper has many free electrons that are loosely held and can move easily when a voltage is applied ✓✓; rubber has almost no free electrons – its electrons are tightly bound, so current cannot flow ✓✓. (4)

**2.4** Any FOUR stages in order ✓✓✓✓: generation at the power station (about 22 kV); step-up transformer to very high voltage (e.g. 400 kV); transmission lines (national grid); substation steps the voltage down; distribution lines; pole / mini-substation transformer to 230 V; service connection and meter to the house (4)

**[16]**

## QUESTION 3

**3.1.1** $\mathrm{I} = \mathrm{P} \div \mathrm{V}$ ✓ $= 40 \div 230 = 0{,}17\ \mathrm{A}$ ✓ (2)

**3.1.2** $\mathrm{R} = \mathrm{V}^2 \div \mathrm{P}$ ✓ $= 230^2 \div 40 = 1\,322{,}5\ \Omega$ ✓ $(\text{or } \mathrm{V} \div \mathrm{I})$ (2)

**3.1.3** $\mathrm{W} = \mathrm{P} \times \mathrm{t}$ ✓ $= 40 \times 600\ \mathrm{s}$ ✓ $= 24\,000\ \mathrm{J}$ ✓ (3)

**3.1.4** $\mathrm{E} = 0{,}04\ \mathrm{kW} \times 30\ \mathrm{h} = 1{,}2\ \mathrm{kWh}$ ✓✓; $\text{cost} = 1{,}2 \times \mathrm{R}3{,}20 = \mathrm{R}3{,}84$ ✓ (3)

**3.2.1** $\mathrm{R_P} = (12 \times 4) \div (12 + 4)$ ✓ $= 3\ \Omega$ ✓; $\mathrm{R_T} = 6 + 3$ ✓ $= 9\ \Omega$ ✓ (4)

**3.2.2** $\mathrm{I} = \mathrm{V} \div \mathrm{R}$ ✓ $= 9 \div 9 = 1\ \mathrm{A}$ ✓ (2)

**3.2.3** $V_P = I \times R_P$ ✓ $= 1 \times 3 = 3$ V ✓ (2)

**[18]**

## QUESTION 4

**4.1** (a) yellow ✓, violet, brown, gold ✓ (b) red, red ✓, orange, silver ✓ (4)

**4.2** (a) Series: $V = 6 \times 1{,}5 = 9$ V ✓, $I = 0{,}5$ A ✓✓ (b) each group: $3 \times 1{,}5 = 4{,}5$ V ✓; two groups in parallel: $V = 4{,}5$ V, $I = 2 \times 0{,}5 = 1$ A ✓✓ (6)

**4.3** Pure silicon is doped (impurities added) ✓. N-type: a pentavalent impurity (5 valence electrons, e.g. phosphorus, arsenic) adds free electrons ✓✓. P-type: a trivalent impurity (3 valence electrons, e.g. boron, gallium) creates holes ✓ (4)

**4.4** Correct symbol (triangle pointing to a bar, anode and cathode) ✓. Forward bias: anode more positive than cathode, the diode conducts (after about 0,7 V) ✓✓. Reverse bias: cathode more positive, the diode blocks current ✓ (4)

**[18]**

## QUESTION 5

**5.1** (a) fuse – rectangle with a line through it ✓ (b) LED – diode symbol with two arrows pointing away ✓ (c) electrolytic capacitor – one straight and one curved / open plate with + sign ✓ (d) switch – a break in the line with a hinged contact ✓ (4)

**5.2** Any SIX in order ✓✓✓✓✓✓: clean the board and component leads; insert the component through the holes; heat the copper strip and the lead together with a clean, tinned tip; feed solder onto the joint (not the iron) until it flows; remove the solder, then the iron; let the joint cool without moving it; trim the excess lead; inspect for a shiny cone-shaped joint and no bridges (6)

**5.3** Any TWO ✓✓: keep the iron in its stand; work in a ventilated area (fumes); wear safety glasses; do not touch the tip; switch off and unplug after use; wash hands (lead solder) (2)

**5.4** Read the colour code to find the nominal value and tolerance ✓; switch the multimeter to the correct resistance range ✓; make sure the resistor is not in a powered circuit / remove one leg ✓; measure without touching both leads with your fingers ✓; calculate the allowed range (e.g. 470 Ω ± 5% = 446,5 Ω to 493,5 Ω) ✓; the resistor is good if the reading is inside the range ✓ (6)

**[18]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**